

A Sparse-Defect Counterexample to Abelian-Border Periodicity

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Abstract

Charlier, Harju, Puzynina, and Zamboni asked whether an infinite word whose sufficiently long factors are all Abelian bordered must be Abelian periodic. We answer the question negatively, already over the binary alphabet. The construction starts from a four-height periodic nearest-neighbor path whose every anti-diagonal contains all parity-compatible endpoint sums. Sparse finite replacement walks then destroy every constant arithmetic-progression tail while a five-witness redundancy preserves the full local anti-diagonal palette. The resulting binary word is not Abelian ultimately periodic, although every factor of length at least 141 has an Abelian border. The finite periodic-palette and replacement certificates are exact and independently reproducible.

Keywords. Abelian border; infinite word; Abelian periodicity; counterexample; combinatorics on words; sparse defects.

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1 Introduction

A finite word is Abelian bordered if it has a nonempty proper prefix Abelian equivalent to a suffix. Charlier, Harju, Puzynina, and Zamboni [1, Question 1] asked whether an infinite word for which every sufficiently long factor is Abelian bordered must be Abelian periodic. Their nonperiodic examples retained Abelian periodicity. Fici and Puzynina later recorded the same question as Problem 42 in their survey [2].

Previous work by the present author determined the first finite-threshold layers: if every Abelian-unbordered factor has length at most 14, then the word is Abelian ultimately periodic with an Abelian period of length at most 14, and this bound is sharp [3]. Those results did not decide what happens under an arbitrary finite bound. The present construction shows that the unrestricted statement is false.

The mechanism has two parts. First, a periodic height path on $\{0, 1, 2, 3\}$ supplies every parity-compatible height sum near every center. Second, widely separated replacement blocks are placed so that every arithmetic progression is changed infinitely often. Five translated witnesses for each center and sum ensure that a single replacement block cannot destroy them all.

The result is not a statement about the minimum possible threshold, period, or height range. The constants 28 and 141 are explicit parameters of one construction. Recent work on mutually Abelian-bordered finite words [4] and on Abelian periodicity of purely morphic words [5] concerns different object classes.

2 Definitions and the height-path translation

Two finite words are *Abelian equivalent* if they have the same Parikh vector; in particular, Abelian-equivalent words have the same length. For binary words of the same length, this is equivalent to containing the same number of 1s. A finite word is *Abelian bordered* if it has a nonempty proper prefix Abelian equivalent to a suffix. An infinite word w is *Abelian ultimately periodic* if

$$w = uv_0v_1v_2\cdots,$$

where the nonempty finite words v_n are pairwise Abelian equivalent. Following [1, 2], we use *Abelian periodic* with this allowed finite preperiod.

For a binary word $w = w_0w_1\cdots$, define

$$z_0 = 0, \quad z_{n+1} - z_n = 2w_n - 1. \quad (2.1)$$

Then z is an integer nearest-neighbor path and

$$|w[u, v)]_1 = \frac{v - u + z_v - z_u}{2}. \quad (2.2)$$

Lemma 2.1 (Center contraction). *Fix $C \geq 1$. Every factor of w of length at least C is Abelian bordered if and only if, for all $0 \leq i < j$, there exist $i \leq a \leq b \leq j$ such that*

$$a + b = i + j, \quad b - a < C, \quad z_a + z_b = z_i + z_j. \quad (2.3)$$

Proof. If $j - i < C$, take $(a, b) = (i, j)$. Suppose $j - i \geq C$. An Abelian border can be chosen with length $1 \leq k \leq (j - i)/2$: if a longer prefix and suffix overlap, deleting their common overlap gives the complementary shorter border. By (2.2), equality of the two Parikh vectors is equivalent to

$$z_{i+k} + z_{j-k} = z_i + z_j.$$

Replace (i, j) by $(i + k, j - k)$ and repeat while the new gap is at least C . The gap strictly decreases, whereas the center and endpoint sum remain fixed, so (2.3) follows.

Conversely, if $j - i \geq C$, a pair supplied by (2.3) cannot be the original endpoint pair. Writing $k = a - i = j - b > 0$ and applying (2.2) gives an Abelian border of length k . $\square$

Lemma 2.2 (Arithmetic-progression criterion). *Assume that z is bounded. The word w is Abelian ultimately periodic if and only if there exist $A \geq 0$ and $h \geq 1$ such that*

$$z_{A+nh} \quad (n \geq 0)$$

is constant.

Proof. If the length- h blocks beginning at $A + nh$ have one common Parikh vector, (2.2) says that

$$z_{A+(n+1)h} - z_{A+nh} = \delta$$

is independent of n . Boundedness forces $\delta = 0$. Conversely, a constant arithmetic-progression tail makes all those length- h blocks have the same number of 1s by (2.2). $\square$

3 A periodic full-palette skeleton

Set $d = 28$, and extend the height word

$$P = 0101010101232323210123232321 \quad (3.1)$$

periodically to $p: \mathbb{N}_0 \rightarrow \{0, 1, 2, 3\}$. Consecutive heights, including the last and first entries of P , differ by 1.

For a center residue $m \in \mathbb{Z}/d\mathbb{Z}$, define

$$\Pi_m = \{p_r + p_{m-r} : r \in \mathbb{Z}/d\mathbb{Z}\}. \quad (3.2)$$

Lemma 3.1 (Full palette). *For every $m \in \mathbb{Z}/d\mathbb{Z}$,*

$$\Pi_m = \begin{cases} \{0, 2, 4, 6\}, & m \equiv 0 \pmod{2}, \\ \{1, 3, 5\}, & m \equiv 1 \pmod{2}. \end{cases} \quad (3.3)$$

Proof. Substitution of the 28 entries of (3.1) into (3.2) gives (3.3). The exact verification program checks equality, rather than mere containment, for all 28 center residues. $\square$

We also need to change the skeleton at an arbitrary residue without breaking the nearest-neighbor path at the boundary of one period block.

Lemma 3.2 (Replacement walks). *For every target residue $t \in \mathbb{Z}/d\mathbb{Z}$, there exist a start residue s_t , an offset $1 \leq r_t < d$, and a word $Q^{(t)} = q_0 q_1 \cdots q_{d-1}$ over $\{0, 1, 2, 3\}$ such that*

1. $t \equiv s_t + r_t \pmod{d}$;
2. $q_0 = p_{s_t}$ and $|q_{d-1} - q_0| = 1$;
3. $|q_{j+1} - q_j| = 1$ for $0 \leq j < d - 1$;
4. $q_{r_t} \neq p_t$.

Proof. The 28 explicit certificates are listed in Appendix A. The supplementary verifier checks them directly; its deterministic recursive depth-first search also regenerates one valid certificate for every target residue. Conditions 2 and 3 ensure that replacing one aligned period block by $Q^{(t)}$ preserves the nearest-neighbor condition both inside the block and across its two boundaries. $\square$

4 Sparse defects

Enumerate the pairs $(a, h) \in \mathbb{N}_0 \times \mathbb{N}$ in a sequence $(a_j, h_j)_{j \geq 1}$ in which every pair occurs infinitely many times. At stage j , choose a target

$$t_j \in \{a_j + nh_j : n \geq 0\}$$

far enough to the right that the aligned replacement interval I_j supplied by Lemma 3.2 lies after all previous intervals, the distance between successive intervals is greater than $6d$, and these distances tend to infinity. This is possible because every arithmetic progression is unbounded and the target offset inside a replacement block is less than d .

On I_j , replace p by $Q^{(t_j \bmod d)}$; outside all replacement intervals retain p . Lemma 3.2 gives a nearest-neighbor path

$$z: \mathbb{N}_0 \longrightarrow \{0, 1, 2, 3\}$$

such that $z_{t_j} \neq p_{t_j}$ for every j . Let D be the set of positions where $z \neq p$.

Lemma 4.1. *The set D has natural density zero, and every arithmetic progression $\{a + nh : n \geq 0\}$ contains infinitely many designated targets t_j .*

Proof. The second assertion follows from the repeated enumeration. The intervals have one fixed length and their successive gaps tend to infinity. Given $G > 0$, all but finitely many consecutive intervals are separated by at least G ; hence their upper density is at most $d/(d + G)$. Letting $G \rightarrow \infty$ proves that the interval union, and therefore D , has density zero. $\square$

5 Preservation of center contraction

Lemma 5.1. *The path z satisfies (2.3) with $C = 141$.*

Proof. Let $0 \leq i < j$ with $j - i \geq 141$, set $N = i + j$, and put $v = z_i + z_j$. Because every nearest-neighbor path has fixed height parity, v is one of the values allowed by the corresponding case of (3.3). Choose $r \pmod{d}$ such that

$$p_r + p_{N-r} = v.$$

Among the pairs (a, b) satisfying $a + b = N$ and $a \equiv r \pmod{d}$, choose (a_0, b_0) with $|a_0 - b_0| \leq d$. Consider the five pairs

$$(a_k, b_k) = (a_0 + kd, b_0 - kd), \quad -2 \leq k \leq 2. \quad (5.1)$$

They have the same center and the same skeleton sum v , and

$$|a_k - b_k| \leq 5d = 140. \quad (5.2)$$

Since $N \geq 141$, all endpoints in (5.1) are nonnegative; exchanging the two entries of a pair if necessary does not change its properties.

The candidate endpoints occupy a band of span at most $5d$. Since distinct replacement intervals are separated by more than $6d$, this band meets at most one of them. A replacement interval of length d meets at most two members of the progression (a_k) and at most two members of (b_k) . It therefore contaminates at most four of the five pairs. For a remaining clean pair,

$$z_{a_k} + z_{b_k} = p_{a_k} + p_{b_k} = v = z_i + z_j.$$

Together with (5.2), this is (2.3). Because the pair has center $N/2$, its gap is smaller than the original gap, so it lies inside $[i, j]$. $\square$

6 Destruction of every Abelian period

Lemma 6.1. *The path z is not constant on the tail of any arithmetic progression.*

Proof. Fix $A \geq 0$ and $h \geq 1$. The periodic sequence p_{A+nh} is either constant or takes at least two values periodically.

In the second case, each of two distinct values occurs on a subprogression of positive natural density in $\mathbb{N}_0$. A zero-density set cannot contain a tail of either subprogression. Hence z_{A+nh} takes both values infinitely often outside D .

In the first case, the repeated enumeration designates infinitely many distinct targets on the progression, and each target is changed from its skeleton value. The progression also contains infinitely many points outside D , because it has positive density while D has density zero. Thus z_{A+nh} is again not eventually constant. $\square$

7 Main theorem

Theorem 7.1. *There exists a binary infinite word that is not Abelian ultimately periodic but whose every factor of length at least 141 is Abelian bordered. Consequently, the answer to Question 1 of [1] and Problem 42 of [2] is negative.*

Proof. Define

$$w_n = \frac{1 + z_{n+1} - z_n}{2}.$$

The nearest-neighbor property makes $w_n \in \{0, 1\}$. Lemmas 5.1 and 2.1 show that every factor of length at least 141 is Abelian bordered. Lemmas 6.1 and 2.2 show that w is not Abelian ultimately periodic. $\square$

8 Remarks

1. Theorem 7.1 does not assert that 141 is the least possible threshold.
2. The period 28 and the four-height range are not claimed to be minimal.
3. The finite verification discovers and checks the skeleton certificates, but the infinite sparse-set and arithmetic-progression arguments are mathematical proofs independent of finite-prefix testing.
4. No morphic representation of the construction is asserted. Reference [5] treats the restricted class of purely morphic words and does not cover the unrestricted hypothesis of Theorem 7.1.

A Replacement certificates

Each row gives $(t, s_t, r_t, Q^{(t)})$. The word has length 28.

t	s_t	r_t	$Q^{(t)}$
0	1	27	1010101010101010101010101012
1	3	26	101010101010101010101010101232
2	0	2	012101010101010101010101010101
3	0	3	012321010101010101010101010101
4	0	4	010121010101010101010101010101
5	0	5	010123210101010101010101010101
6	0	6	010101210101010101010101010101
7	0	7	010101232101010101010101010101
8	0	8	010101012101010101010101010101
9	0	9	010101012321010101010101010101
10	0	10	010101010101010101010101010101
11	0	11	010101010101010101010101010101
12	0	12	010101010101010101010101010101
13	0	13	010101010101010101010101010101
14	0	14	010101010101010101010101010101
15	0	15	010101010101010101010101010101
16	0	16	010101010101010101010101010101
17	0	17	010101010101010101232101010101
18	0	18	0101010101010101012101010101
19	0	19	0101010101010101012321010101
20	0	20	010101010101010101010101010101
21	0	21	010101010101010101010101010101
22	0	22	010101010101010101010101010101
23	0	23	010101010101010101010101010101
24	0	24	010101010101010101010101010101

t	s_t	r_t	$Q^{(t)}$
25	0	25	010101010101010101010101010101
26	0	26	010101010101010101010101010101
27	1	26	10101010101010101010101010101232

B Reproducibility

The supplementary file `verify_sparse_palette_counterexample.py` has SHA-256

92628A23B7ACED4E111A9909C0EBC429
07981737479AB902240CABC0B61B7ECC.

The program checks the cyclic nearest-neighbor skeleton, all 28 palette equalities, all 28 replacement certificates, the four-of-five contamination bound, and a finite sparse-prefix positive control. Its role and limitations are stated in the program header.

AI-assisted research and writing disclosure

The author used OpenAI Codex for computer-assisted exploration, exact verification, proof checking, literature triage, and drafting support. The author reviewed the claims and assumes responsibility for the content.

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